Unit 5 · Session 1 · Tue, Nov 17, 2026

# Renal Pharmacotherapeutics — Session 1

3 parts · every fact cited to its slide or textbook page. About 120 minutes of focused study.

## Study plan

**20 min**
**Read the guide**
One pass, top to bottom. Bold = what the lecture emphasized.

**50 min**
**Diagrams & tables**
Redraw each flow from memory; cover a table column and recall it.

**15 min**
**Cram sheet**
Hide answers and quiz yourself. Repeat until it's automatic.

**30 min**
**Self-test**
Answer before you look. Read every explanation, even when right.

**5 min**
**Objectives check**
Tick each objective only if you can teach it out loud.

**HOW TO READ THIS GUIDE**

- **Bold** = emphasized on the lecture slides
- TB textbook only · SG course study guide only
- S12 = slide 12 · S12 (notes) = speaker notes · SG U1-R3 = study guide item · p. 17 = Katzung page
- Open **Sidekick** (bottom right) to search, listen, drill flashcards, quiz, or chat about this lecture

# The whole lecture in 6 sentences

1. Falling **GFR** makes renally cleared drugs **accumulate**: check CrCl first, then lower the maintenance dose, extend the interval, or both (loading doses often unchanged). S5; S6; S8; SG U5-R1

2. Each diuretic acts at one segment: **CAI/osmotic** proximal tubule, **loop** thick ascending limb, **thiazide** distal tubule, **$K^+$-sparing** collecting duct. S7; SG U5-R2; SG U5-M

3. **Loops** (NKCC2, ~25%) are most potent and **raise** urinary calcium; **thiazides** (NCC, ~5–8%) are first-line for hypertension and **lower** it. S10; S12; S14; SG U5-R3

4. Thiazides fade below **GFR ~30** — switch to a loop. S15; SG U5-R4

5. **Spironolactone/eplerenone** block the aldosterone receptor; **amiloride/triamterene** block ENaC; all risk **hyperkalemia**, worse with ACEi/ARB or CKD. S17; S23; SG U5-R5

6. **Acetazolamide**: glaucoma, altitude sickness, alkalosis (causes acidosis). **Mannitol**: raised ICP/IOP; avoid in heart failure. S19; S20; SG U5-R6

# Learning objectives

IO = syllabus unit objective · SO = this session's objectives slide.

**IO5.1** Identify major pharmacologic classes used in the management of renal conditions and describe their mechanisms of action.

Part 1 · Renal Dosing & Diuretic Sites
Part 2 · Loop & Thiazide Diuretics
Part 3 · Potassium-Sparing, CAI & Osmotic Diuretics

**IO5.2** Describe the indications, contraindications, adverse effects, and clinically significant interactions associated with medications used to treat renal conditions.

Part 2 · Loop & Thiazide Diuretics
Part 3 · Potassium-Sparing, CAI & Osmotic Diuretics

**IO5.3** Compare pharmacologic agents within and across therapeutic classes to distinguish clinically relevant differences in efficacy, safety, and use.

Part 2 · Loop & Thiazide Diuretics
Part 3 · Potassium-Sparing, CAI & Osmotic Diuretics

**IO5.4** Select appropriate pharmacologic therapy for common renal conditions based on clinical presentation and patient-specific factors.

Part 2 · Loop & Thiazide Diuretics
Part 3 · Potassium-Sparing, CAI & Osmotic Diuretics

**IO5.5** Identify appropriate dosing, administration, and monitoring considerations for medications used in renal conditions.

Part 1 · Renal Dosing & Diuretic Sites

**IO5.6** Explain important medication-use and safety information to patients receiving pharmacotherapy for renal.

Part 3 · Potassium-Sparing, CAI & Osmotic Diuretics

**SO1** Identify the major pharmacologic classes used in the management of renal conditions — diuretics — and describe their mechanisms of action at specific nephron sites.

Part 1 · Renal Dosing & Diuretic Sites
Part 2 · Loop & Thiazide Diuretics
Part 3 · Potassium-Sparing, CAI & Osmotic Diuretics

**SO2** Apply basic pharmacokinetic principles to drug dosing in renal impairment, including creatinine clearance/GFR-based dose and interval adjustment.

Part 1 · Renal Dosing & Diuretic Sites

**SO3** Describe the indications, contraindications, adverse effects, and clinically significant interactions of loop, thiazide, potassium-sparing, carbonic anhydrase inhibitor, and osmotic diuretics.

Part 2 · Loop & Thiazide Diuretics
Part 3 · Potassium-Sparing, CAI & Osmotic Diuretics

**SO4** Compare diuretic classes to distinguish clinically relevant differences in efficacy, electrolyte effects, and clinical use.

Part 2 · Loop & Thiazide Diuretics
Part 3 · Potassium-Sparing, CAI & Osmotic Diuretics

**SO5** Identify dosing, administration, and monitoring considerations, and explain key medication-safety counseling points for patients on diuretic therapy.

Part 2 · Loop & Thiazide Diuretics
Part 3 · Potassium-Sparing, CAI & Osmotic Diuretics

## ABOUT THIS LECTURE

- **PA PHARMACOLOGY COURSE · LECTURE 5 · SESSION 1 OF 2** — Renal Pharmacotherapeutics. **Session 1: Renal Physiology, Dosing Principles, and Diuretic Pharmacology | Tuesday, November 17 | 8:30–11:00 AM** S1
- Two ideas run through this unit. First, the kidney clears many drugs, so renal function changes dosing. Second, every diuretic works at one specific nephron segment, and that location predicts its effects, its adverse effects and its interactions. S1 (notes); S2 (notes)
- Arc of the session: renal physiology and dosing principles first, then the five diuretic classes organized by where they act along the nephron. Preview: antihypertensive therapy in chronic kidney disease and the cardiovascular-renal overlap continue in Session 2 (Thursday, November 19). S2; S2 (notes)
- Session 2 builds directly on this foundation — RAAS-blocking antihypertensives are, mechanistically, the natural next step after today's diuretic content. S2 (notes)
- Readings: Katzung chapters 15 (Diuretic Agents) and 3 (Pharmacokinetics & Pharmacodynamics: Rational Dosing and the Time Course of Drug Action) for today; chapters 11 (Antihypertensive Agents) and 12 (Vasodilators & the Treatment of Angina Pectoris) are supplemental for Session 2, not today. S3; S3 (notes)
- Chapter 11 (Antihypertensive Agents) frames hypertension management in chronic kidney disease; Chapter 12 the cardiovascular-renal overlap with angina pharmacology. Both supplemental chapters are covered in Session 2 (Thursday, November 19), not today. S3; S3 (notes)
- Required text: Katzung's Basic and Clinical Pharmacology, 16th Edition (Todd W. Vanderah, 2024/2025, McGraw-Hill Education; ISBN 1260463303 eBook; AccessMedicine). Pharmacotherapy Principles and Practice (Chisholm-Burns et al., 2022) remains additional support alongside all four chapters. S3
- **Appendix** 1 abbreviation key: PO = by mouth; IV = intravenous; IM = intramuscular; SL = sublingual; BID / TID / QID = two / three / four times daily; ER = extended-release; ICP = intracranial pressure; IOP = intraocular pressure. S26; S26 (notes)
- CKD = chronic kidney disease; AKI = acute kidney injury; GFR = glomerular filtration rate; CrCl = creatinine clearance; ENaC = epithelial sodium channel; ACE = angiotensin-converting enzyme; ARB = angiotensin receptor blocker. S26; S26 (notes)
- MI = myocardial infarction; HF = heart failure; CYP = cytochrome P450; NSAID = nonsteroidal anti-inflammatory drug; CBC = complete blood count. Use these as a study reference. S26; S26 (notes)
- **Appendix 2: DRUG PROFILE · SUPPORTING** — supporting drugs at a glance, listed for context to explain interactions, not doses. S38; S38 (notes)
- The appendix gives **Formulation, dose, indications, adverse effects, contraindications and toxicity** for each drug. For each, know the dose logic, the key adverse effect, the contraindication that changes decisions and the toxicity you would need to recognize. **Standard**

**adult dosing for teaching; verify against current prescribing information.** S26; S26 (notes); S27-37

- **Original figures in this deck:** Figures 1–7 (Renal Dose Adjustment: Dose, Interval, or Both; Diuretic Sites of Action Along the Nephron; The Na-K-2Cl Cotransporter (NKCC2): Loop Diuretic Target; Thiazides: Distal Tubule NCC Block and Calcium Reabsorption; S39
- Collecting Duct: Aldosterone Blockers vs. ENaC Blockers; Proximal Tubule: Acetazolamide and Mannitol; Diuretic Class Comparison: Natriuretic Potency). S39
- Figures 2, 3 and 7 were generated by the course director using the AI tool Claude to illustrate standard, non-copyrighted pharmacologic concepts; they are original diagrams, not reproduced from Katzung's Basic and Clinical Pharmacology or any other copyrighted source. S39; S39 (notes)
- Figures 1, 4, 5 and 6 are illustrative AI-generated schematics of standard mechanisms, not reproduced from any copyrighted source. **Reference:** Anthropic. (2026). Claude (Sonnet 5) [Large language model]. https://claude.ai S39; S39 (notes)

**SLIDE DISCREPANCIES**

- **Thiazide GFR cutoff:** slides teach about 30 mL/min (switch to a loop); the textbook says the exact cutoff is debated. For the exam, use 30. S15; p. 279
- **Hypomagnesemia:** slides list it under loops; the textbook says it is more likely with thiazides. Both classes can cause it. S10; p. 273
- **Chlorthalidone half-life:** profile says 40–60 hours; textbook table says 50–60 hours. Either way, the longest-acting thiazide. S31; p. 281
- **Mannitol dose:** profile 0.25–1 g/kg; textbook 1–2 g/kg. Use the slide dose. S37; p. 275

1 PART 1 · DRUG CLASS

# Renal Dosing & Diuretic Sites

SO1 SO2 IO5.1 IO5.5

**HIGH-YIELD · KNOW THIS COLD**

- Falling GFR → accumulation → toxicity: cut the maintenance dose, stretch the interval, or both; check CrCl first. S6; S8; SG U5-R1
- Sodium reabsorbed: proximal ~65%, loop ~25%, distal 5–8%, collecting duct 2–3%. S7 (notes)

## CLASS

- **Where diuretics act, and why renal function changes how every drug is dosed.** Renal physiology and pharmacokinetic principles set the dosing logic for everything that follows. S4; S4 (notes)
- **The kidney is a major route of elimination for many drugs and active metabolites — renal function directly determines how long a drug stays in the body.** S5
- Glomerular filtration rate (GFR) and creatinine clearance (CrCl) are the standard measures used to estimate renal drug-elimination capacity. S5; S5 (notes)
- This is foundational Chapter 3 content applied specifically to the renal unit: renal dosing is not a separate topic but the lens through which every drug in this unit (and this course) should be viewed. S5 (notes)

## PROTOTYPE + OTHER DRUGS NAMED

- Digoxin, most antibiotics (e.g., vancomycin, many beta-lactams), many direct oral anticoagulants, metformin, and the diuretics covered later in this session are examples of drugs requiring renal dose adjustment. S5; S5 (notes)

## MECHANISM OF ACTION

- Flow (Figure 1): GFR falls → renally cleared drug or active metabolite accumulates → toxicity. S6; S6 (notes)

## INDICATIONS

- Three levers: reduce the dose, extend the dosing interval, or both — the specific approach depends on the drug. Reduced GFR makes a renally cleared drug accumulate if unadjusted. S6; S8; SG U5-R1

- **Figure 2. Original illustrative diagram (not from Katzung): filtrate flows through the proximal tubule, thick ascending limb, distal convoluted tubule, and collecting duct — each diuretic class blocks sodium reabsorption at a different, specific segment.** S7
- Share of filtered sodium reabsorbed: proximal tubule about 65%, thick ascending limb about 25%, distal tubule 5-8%, collecting duct 2-3% — this previews the potency ranking in Figure 7. S7 (notes)
- This nephron map is the conceptual spine of the session — every diuretic class is anchored back to its segment. S7 (notes)
- Match each diuretic class to the nephron segment where it acts: loop diuretics (furosemide) — thick ascending limb; thiazide diuretics (hydrochlorothiazide) — distal convoluted tubule. SG U5-R2; SG U5-M
- Potassium-sparing — aldosterone antagonists (spironolactone) and ENaC blockers (amiloride) — collecting duct; carbonic anhydrase inhibitors (acetazolamide) — proximal tubule; osmotic diuretics (mannitol) — throughout. SG U5-R2; SG U5-M
- CAI — proximal tubule; osmotic — acts osmotically throughout the filtrate-containing segments, mainly the proximal tubule and descending limb. SG U5-R2; SG U5-M
- **TB** Renal tubule transport mechanisms: the proximal tubule reabsorbs about 66% of $Na^+$ and 85% of $NaHCO_3$; organic acid secretion there delivers most diuretics to their luminal site. p. 262; p. 263
- **TB** Collecting tubule system: only 2–5% of NaCl, but where aldosterone acts and $K^+$ is secreted; upstream diuretics deliver more $Na^+$ here — the basis of diuretic $K^+$ wasting. p. 264; p. 265
- Reduce the maintenance dose when the drug's effect is concentration-dependent; extend the dosing interval when the drug has a long half-life; monitor for toxicity after each dose change. S6
- Loading doses are often unchanged; maintenance doses are what get adjusted. S6 (notes); S8

- TB Renal autacoids: adenosine (A1) lowers GFR; prostaglandins (PGE2) aid loop action; natriuretic peptides (nesiritide) and urodilatin cause natriuresis but have a limited clinical role. p. 266

### ADVERSE EFFECTS (BOXED WARNINGS FIRST)

- As GFR falls, renally cleared drugs accumulate — dosing must be adjusted by reducing the dose, extending the interval, or both. S5
- Active or toxic metabolites that are renally cleared matter just as much as the parent drug. S8

### CONTRAINDICATIONS

*Not covered in materials*

### DRUG INTERACTIONS

- **Renal impairment also raises drug-interaction risk — drugs that are individually safe can become dangerous in combination when renal clearance is already reduced (a theme that returns explicitly in Session 2's "triple whammy" discussion).** S5

### PHARMACOKINETICS

- TB Pharmacokinetics in two numbers: volume of distribution links amount in the body to concentration; clearance links elimination rate to concentration. p. 43
- TB Maintenance dose = clearance × target concentration (steady state depends only on clearance); loading dose = volume of distribution × target concentration, given slowly IV. p. 51; p. 52
- TB Drug accumulation: doses repeated more often than every four half-lives accumulate; steady state takes about four half-lives. p. 47; p. 48

### MONITORING

- Check creatinine clearance before dosing any renally cleared drug (lower maintenance dose, extend the interval, or check CrCl) — especially in older adults. S6; S8
- Reassess as renal function changes — acute illness, dehydration, or new nephrotoxic drugs can drop GFR quickly. S8; S8 (notes)

- When in doubt, consult dosing references or clinical pharmacy — this is not a "memorize every number" exercise; the point is the habit of checking renal function before dosing. S8; S8 (notes)
- **TB** Distribution & clearance: adjust a renally cleared drug in proportion to creatinine clearance (Cockcroft-Gault; women 85% of the calculated value). p. 55
- **TB** The target concentration strategy: choose a target, predict V and CL from weight and renal function, dose, measure the response, revise, repeat. p. 53
- **TB** Dosing history: without it a drug level loses its predictive value; draw digoxin at least 6 hours after a dose. p. 54; p. 55

**CLINICAL PEARL / WHY**

- Renal function determines drug dosing broadly, not just for diuretics — GFR/creatinine clearance should be checked before dosing any renally cleared drug (Chapter 3). S25
- **TB** Textbook case: 85-year-old, 60 kg, CrCl 22 mL/min — digoxin clearance falls to 3.7 L/h, so about 127 mcg/day: two 62.5-mcg tablets. p. 42; p. 56

## Step by step

### Dosing a renally cleared drug S6; S8

Estimate CrCl or eGFR → Is the drug (or active metabolite) renally cleared? →

Concentration-dependent → lower the maintenance dose →

Long half-life → extend the interval →

Reassess when illness, dehydration or nephrotoxins change GFR

**Figure 1 · Renal dose adjustment: dose, interval, or both. Schematic (illustrative)** — Figure 1. Reduced GFR slows clearance, so a renally cleared drug accumulates to higher levels (illustrative schematic). Fixes: lower the maintenance dose, extend the interval, or both; loading doses are often unchanged. S6; S6 (notes)

**Figure 2 · Diuretic sites of action along the nephron.** Original illustrative diagram (not from Katzung): filtrate flows left to right through the proximal tubule (~65% of filtered $Na^+$), thick ascending limb (~25%), distal convoluted tubule (~5–8%) and collecting duct (~2–3%); each class blocks sodium reabsorption at its own segment, and the share reabsorbed there predicts its potency. S7; S7 (notes)

## Renal dosing: what changes (S8)

| What changes | Practical consideration |
|---|---|
| **Dose, interval, or both — depends on the drug** | Estimate CrCl or eGFR before dosing, especially in older adults |
| **Loading dose often unchanged; maintenance dose adjusted** | Reassess when illness, dehydration or nephrotoxins drop GFR |
| **Renally cleared active/toxic metabolites count too** | Use dosing references or clinical pharmacy |

How to tell them apart. Sources: S8

**WHY IT HAPPENS**

- Why is the loading dose often unchanged? It fills the volume of distribution, which kidney failure doesn’t shrink much; clearance sets the maintenance dose. S8; p. 51; p. 52

2 PART 2 · DRUG CLASS

# Loop & Thiazide Diuretics

SO1 SO3 SO4 SO5 IO5.1 IO5.2 IO5.3 IO5.4

**HIGH-YIELD · KNOW THIS COLD**

- Loop = NKCC2, thick ascending limb, most potent, ↑ $Ca^{2+}$ excretion, ototoxicity. Thiazide = NCC, DCT, first-line HTN, ↓ $Ca^{2+}$ excretion. S14; SG U5-R3; SG U5-R4
- Thiazides fade below GFR ~30 → switch to a loop. S15; SG U5-R4

## CLASS

- **The two workhorse classes — one for potent natriuresis, one for everyday hypertension control.** They act on the thick ascending limb and the distal tubule. S9; S9 (notes)
- Loops are the most potent class; thiazides are first-line for hypertension. S10; S12

## PROTOTYPE + OTHER DRUGS NAMED

- Furosemide, bumetanide, torsemide: loop diuretics. Hydrochlorothiazide: thiazide; chlorthalidone: thiazide-like. S10; S12

## MECHANISM OF ACTION

- **Mechanism: block the Na-K-2Cl cotransporter (NKCC2) in the thick ascending limb — the site responsible for ~25% of filtered sodium reabsorption.** S10
- **Figure 3. Original illustrative diagram (not from Katzung): loop diuretics block NKCC2 in the thick ascending limb, reducing Na+/Cl-/K+ reabsorption and collapsing the lumen-positive potential that normally drives paracellular calcium and magnesium reabsorption —** S11
- **explaining the increased calcium and magnesium excretion seen with this class.** The lumen-positive potential (from $K^+$ recycling) is the link students skip when they memorize “loops cause hypocalcemia” without the why. S11; S11 (notes); SG U5-R3

## INDICATIONS

- Loops: edema (heart failure, cirrhosis, renal disease) and, at times, hypertension when a thiazide alone is insufficient (e.g., significant renal impairment). S10
- Loop diuretics increase calcium excretion — useful in hypercalcemia, and the opposite effect of thiazides. S10; S10 (notes)
- Hydrochlorothiazide, chlorthalidone — first-line agents for essential hypertension; chlorthalidone has a longer half-life and is often preferred for 24-hour BP control. S12
- Thiazides decrease calcium excretion — can be therapeutically useful in patients with osteoporosis and recurrent calcium-containing kidney stones. S12; S13
- TB Hypercalcemia: normal saline plus IV furosemide (80–120 mg) — a loop alone causes volume contraction and stops working. p. 280

- **Mechanism: block the Na-Cl cotransporter in the distal convoluted tubule — a less potent natriuretic site (~5–8% of filtered sodium) than the loop.** S12
- Follow the thiazide chain step by step (Figure 4): NCC block in the DCT → sodium and water loss → lower intracellular sodium → calcium retained through the NCX exchanger → falling urinary calcium. S13; S13 (notes); SG U5-R3
- Contrast this with loop diuretics, which raise urinary calcium (Figure 3). S13 (notes)
- In one line: loop diuretics collapse the lumen-positive potential driving paracellular $Ca^{2+}$/$Mg^{2+}$ reabsorption (↑ excretion); thiazides increase distal calcium reabsorption through a separate mechanism (↓ excretion). SG U5-R3
- **TB** Loops work from the lumen, so their effect tracks proximal secretion; NSAIDs and probenecid compete and reduce it. NSAIDs also blunt the PGE2 contribution. p. 269; p. 270
- **TB** Nephrolithiasis: thiazide plus more fluid and less salt; do not restrict dietary calcium. p. 280
- **TB** Diabetes insipidus (nephrogenic, including lithium-induced): thiazides reduce polyuria; amiloride blocks lithium entry; monitor lithium levels. p. 280
- **TB** Nonedematous states, hypertension: chlorthalidone lowers BP more than HCTZ but has more adverse effects; loops reserved for GFR < 30–40 mL/min or heart failure. p. 280

### ADVERSE EFFECTS (BOXED WARNINGS FIRST)

- **Adverse effects: hypokalemia, hypomagnesemia, metabolic alkalosis, ototoxicity (especially with rapid IV administration or high doses), and hyperuricemia.** S10
- **Adverse effects: hyponatremia (can be significant, especially in older adults), hypokalemia, hyperglycemia, hyperuricemia (can precipitate gout), and hypercalcemia.** S12
- Loops carry greater risk of ototoxicity and more pronounced electrolyte shifts than thiazides. S14
- Thiazides: hyponatremia, gout, and hypokalemia are common adverse effects. S13

### CONTRAINDICATIONS

- Furosemide carries a sulfonamide-related structure — historical cross-reactivity concern with sulfa allergy, though the clinical risk is now considered low; still worth a careful allergy history. S10; S10 (notes)
- Thiazides lose efficacy as GFR falls below approximately 30 mL/min — loop diuretics are generally preferred once renal function is significantly reduced. S12; S13

- **TB** Thiazide hyperglycemia occurs at HCTZ > 50 mg/d, not at 12.5 mg/d or less; correcting hypokalemia partly reverses it. p. 272
- **TB** Thiazides are the diuretics most associated with gout and cause hypomagnesemia more than loops; loop ototoxicity is usually reversible. p. 271; p. 272; p. 273
- **TB** Risk of hyponatremia with different diuretic classes: thiazides highest (elderly women); loops protective (odds ratio 0.61). p. 272; p. 281

## DRUG INTERACTIONS

- Digoxin: hypokalemia from loops increases toxicity. Aminoglycosides: additive ototoxicity with loops. S38; S38 (notes)
- Lithium: levels rise with thiazides. NSAIDs blunt the diuretic effect and raise AKI risk. S38; S38 (notes)
- **TB** Diuretic combinations: loop agents and thiazides (often metolazone) are synergistic in loop resistance — but cause large fluid shifts and $K^+$ wasting. p. 277

## PHARMACOKINETICS

- Chlorthalidone: long half-life of 40-60 hours. S31
- Torsemide: longer half-life and more predictable absorption than furosemide. Bumetanide is about 40 times more potent than furosemide mg for mg. S28; S29
- **TB** Furosemide lasts 2–3 hours; torsemide 4–6 hours and absorbs in 1 hour, nearly as completely as IV. Torsemide is mainly cleared by the liver. p. 269
- **TB** Chemistry & pharmacokinetics: all thiazides are sulfonamides given orally; chlorothiazide is the only parenteral thiazide. p. 271

## MONITORING

- Loops: monitor potassium, magnesium, creatinine. Thiazides: monitor sodium, potassium, calcium, glucose. S27; S30
- IV furosemide slowly (no faster than 4 mg/min) — ototoxicity with rapid IV or high doses. S27

## CLINICAL PEARL / WHY

- The calcium-handling difference between loops and thiazides is one of the highest-yield comparison points in this whole unit: loops increase Ca2+ excretion, thiazides decrease it — state it both ways. S10 (notes); S25; SG U5-R3

- Case: hypertension and worsening CKD (eGFR now 25 mL/min), well controlled on chlorthalidone for years, but blood pressure has crept up. **What is the most likely explanation, and what change in therapy would you consider?** S15
- **Thiazide diuretics lose much of their natriuretic efficacy once GFR falls below roughly 30 mL/min, because there is too little filtered sodium reaching the distal tubule for the drug to act on meaningfully.** S15
- **A loop diuretic, which acts upstream at the thick ascending limb, remains effective even in advanced CKD and is the more appropriate choice here.** S15; S15 (notes)
- The "thiazides lose efficacy in advanced CKD" point bridges directly to Session 2's hypertension-in-CKD content. S12 (notes)
- **TB** Edematous states, heart failure: loops are standard, but overdiuresis cuts preload and cardiac output; hypokalemia worsens arrhythmias and digitalis toxicity. p. 278
- **TB** Kidney disease and renal failure: little benefit below GFR 5 mL/min; high-dose loops or loop plus metolazone treat overload; avoid acetazolamide. p. 278; p. 279
- **TB** Textbook case: HCTZ added to chronic furosemide in CKD → 15 kg lost, BP 91/58, creatinine 10.8 — hypovolemic AKI; monitor closely, ideally inpatient. p. 261; p. 285

### Step by step

**Loop or thiazide?** S10; S12; S14; S15; p. 280

Edema or advanced CKD (GFR < 30)? →

Yes → loop (furosemide, bumetanide, torsemide) →

Essential hypertension, preserved renal function → thiazide →

Calcium stones or osteoporosis → thiazide helps →

Hypercalcemia → loop (with saline)

**Figure 3 · NKCC2: the loop diuretic target.** Original illustrative diagram (not from Katzung): blocking NKCC2 in the thick ascending limb cuts $Na^+/K^+/2Cl^-$ reabsorption and collapses the lumen-positive potential (from $K^+$ recycling) that drives paracellular $Ca^{2+}$ and $Mg^{2+}$ reabsorption — so loops raise urinary calcium and magnesium. S11; S11 (notes)

**Figure 4 · Thiazides: NCC block and calcium reabsorption. Schematic (illustrative)** — Figure 4. Distal convoluted tubule: thiazide block of NCC lowers intracellular sodium, which favors $Na^+/Ca^{2+}$ exchange (NCX), so more calcium is reabsorbed and urinary calcium falls (illustrative schematic). S13; S13 (notes)

## Loop vs. thiazide diuretics (S14)

| | Loop diuretics | Thiazide diuretics |
|---|---|---|
| **Site, potency** | Thick ascending limb — most potent natriuresis (~25%) | Distal convoluted tubule — modest natriuresis (~5–8%) |
| **Calcium** | Increase $Ca^{2+}$ excretion | Decrease $Ca^{2+}$ excretion |
| **Best use** | Preferred in edematous states and advanced CKD | First-line for essential hypertension with preserved renal function |
| **Watch for** | Greater risk of ototoxicity, more pronounced electrolyte shifts | Lose efficacy as GFR falls below ~30 mL/min |

How to tell them apart. Sources: S14; SG U5-R4

## Loop diuretic profiles: formulation, dose, indications

| Drug | Class | Formulation | Adult dose | Indications |
|---|---|---|---|---|
| **Furosemide (Lasix)** | **Loop diuretic** | Tablets 20/40/80 mg; oral solution; IV injection 10 mg/mL | Edema: 20-80 mg PO once or BID, titrate to response (up to 600 mg/day in severe CKD or HF); IV 20-40 mg slowly (no faster than 4 mg/min) | Edema (heart failure, cirrhosis, kidney disease), hypertension, acute pulmonary edema, hypercalcemia |
| **Bumetanide (Bumex)** | **Loop diuretic** | Tablets 0.5/1/2 mg; IV injection 0.25 mg/mL | Edema: 0.5-2 mg PO once or BID (maximum 10 mg/day); IV 0.5-1 mg | Edema in heart failure, hepatic and renal disease |
| **Torsemide (Demadex)** | **Loop diuretic** | Tablets 5/10/20/100 mg | Edema: 10-20 mg PO daily (up to 200 mg/day); hypertension: 5 mg daily | Edema in heart failure, kidney disease, cirrhosis; hypertension |

How to tell them apart. Sources: S27; S28; S29

## Loop diuretic profiles: adverse effects, contraindications, toxicity

| Drug | Adverse effects | Contraindications | Toxicity / overdose |
|---|---|---|---|
| **Furosemide (Lasix)** | Hypokalemia, hypomagnesemia, metabolic alkalosis, volume depletion, hyperuricemia, ototoxicity | Anuria, hypersensitivity; use caution with sulfonamide allergy (low cross-reactivity), hepatic coma | Ototoxicity with rapid IV or high doses, severe volume and electrolyte depletion, AKI; monitor potassium, magnesium, creatinine |
| **Bumetanide (Bumex)** | Hypokalemia, hyponatremia, hypomagnesemia, muscle cramps, dizziness, ototoxicity | Anuria, hepatic coma, severe electrolyte depletion, hypersensitivity | About 40 times more potent than furosemide mg for mg; profound diuresis, hypokalemia, ototoxicity |

| Drug | Adverse effects | Contraindications | Toxicity / overdose |
|---|---|---|---|
| **Torsemide (Demadex)** | Hypokalemia, dizziness, headache, hyperuricemia, ototoxicity (less than furosemide) | Anuria, hypersensitivity (sulfonylurea or sulfonamide allergy caution) | Volume depletion, hypokalemia, hypotension; longer half-life and more predictable absorption than furosemide |

How to tell them apart. Sources: S27; S28; S29

## Thiazide profiles: formulation, dose, indications

| Drug | Class | Formulation | Adult dose | Indications |
|---|---|---|---|---|
| **Hydrochlorothiazide (Microzide)** | **Thiazide diuretic** | Tablets 12.5/25/50 mg; capsules 12.5 mg; many fixed combinations | Hypertension: 12.5-25 mg PO daily (maximum 50 mg); edema: 25-100 mg daily | First-line hypertension, mild edema, calcium kidney stones, nephrogenic diabetes insipidus |
| **Chlorthalidone (Thalitone)** | **Thiazide-like diuretic** | Tablets 12.5/25/50 mg | Hypertension: 12.5-25 mg PO daily (maximum 50 mg); edema: 50-100 mg daily | Hypertension (longer duration, often preferred), edema |

How to tell them apart. Sources: S30; S31

## Thiazide profiles: adverse effects, contraindications, toxicity

| Drug | Adverse effects | Contraindications | Toxicity / overdose |
|---|---|---|---|
| **Hydrochlorothiazide (Microzide)** | Hyponatremia, hypokalemia, hypercalcemia, hyperglycemia, hyperuricemia and gout, photosensitivity | Anuria, sulfonamide allergy; less effective when GFR is below about 30 mL/min | Severe hyponatremia and hypokalemia, volume depletion, hypercalcemia; monitor sodium, potassium, calcium, glucose |
| **Chlorthalidone (Thalitone)** | Hypokalemia, hyponatremia, hyperuricemia, hyperglycemia, erectile dysfunction | Anuria, sulfonamide allergy, severe hypokalemia | Profound hypokalemia and hyponatremia (more than hydrochlorothiazide at equal dose); long half-life of 40-60 hours |

How to tell them apart. Sources: S30; S31

**SLIDE DISCREPANCY**

- **Thiazide efficacy in advanced CKD**
- Slides: Thiazides lose efficacy below GFR ~30 mL/min; switch to a loop. S12; S15
- Textbook: The exact GFR cutoff is debated; chlorthalidone still lowered BP in stage 4 CKD. p. 279; p. 280

**SLIDE DISCREPANCY**

- **Hypomagnesemia**
- Slides: Listed as a loop adverse effect. S10
- Textbook: More likely with thiazides than with loops. p. 273

**WHY IT HAPPENS**

- Why do thiazides fail at eGFR 25? Too little filtered sodium reaches the distal tubule; a loop acts upstream where the sodium still is. S15

**MNEMONIC (MEMORY AID)**

- Loops Lose calcium; Thiazides Take it back. S10; S12; SG U5-R3

3 PART 3 · DRUG CLASS

# Potassium-Sparing, CAI & Osmotic Diuretics

SO1 SO3 SO4 SO5 IO5.1 IO5.2 IO5.3 IO5.4 IO5.6

**HIGH-YIELD · KNOW THIS COLD**

- Potassium-sparing: aldosterone antagonists (spironolactone, eplerenone) and ENaC blockers (amiloride, triamterene); hyperkalemia worsens with ACEi/ARB or CKD. S17; SG U5-R5
- Acetazolamide = glaucoma, altitude sickness, alkalosis; mannitol = ICP/IOP. S19; SG U5-R6

## CLASS

- **Smaller in potency, but each with a distinct and important clinical niche.** Potassium-sparing, carbonic anhydrase inhibitor and osmotic diuretics act on the collecting duct and the proximal tubule. S16; S16 (notes)
- **Two mechanistically distinct subgroups acting at the collecting duct, both reducing potassium and hydrogen ion secretion.** S17

## PROTOTYPE + OTHER DRUGS NAMED

- Spironolactone, eplerenone: aldosterone receptor antagonists. Amiloride, triamterene: epithelial sodium channel (ENaC) blockers. S17
- Carbonic anhydrase inhibitor: acetazolamide. Osmotic diuretic: mannitol. S19

## MECHANISM OF ACTION

- Aldosterone binds the mineralocorticoid receptor → increases ENaC and Na/K pump activity → K+ and H+ secreted into the tubule. Blocking either step reduces potassium and hydrogen secretion. S18; S18 (notes)
- Spironolactone and eplerenone block the receptor (block aldosterone's effect directly); amiloride and triamterene block ENaC directly, independent of aldosterone. S17; S18; S18 (notes)
- **Acetazolamide (carbonic anhydrase inhibitor):** blocks Na+/HCO3- reabsorption in the proximal tubule → less HCO3- reabsorbed → bicarbonate diuresis, metabolic acidosis; weak diuretic effect. S19; S20; SG U5-R6

## INDICATIONS

- Potassium-sparing agents are often used in combination with a loop or thiazide diuretic specifically to offset potassium loss from the more potent class. S17
- Acetazolamide: valued for non-diuretic uses — glaucoma (reduces aqueous humor production), altitude sickness prophylaxis, and metabolic alkalosis correction; also raised intracranial pressure. S19; S20
- Mannitol: used to reduce elevated intracranial or intraocular pressure and in certain cases of rhabdomyolysis-associated kidney injury prevention. S19
- Know the niche, not an everyday use: acetazolamide = glaucoma/altitude sickness; mannitol = ICP/IOP reduction. S19 (notes)

- **Mannitol (osmotic diuretic):** an inert, filtered but poorly reabsorbed solute that holds water in the tubule lumen osmotically — osmotic diuresis. S19; S20
- Side by side: acetazolamide removes bicarbonate reabsorption; mannitol removes water reabsorption by staying in the lumen. Both act before the loop, so their natriuretic potency is modest. S20 (notes)
- Figure 6 (**Schematic (illustrative)**): proximal tubule — carbonic anhydrase inhibition and osmotic holding of water (illustrative schematic). S20
- **Figure 7. Original illustrative comparison (illustrative values, not from Katzung): loop diuretics block roughly 25% of filtered sodium reabsorption, substantially more than thiazides, carbonic anhydrase inhibitors, or potassium-sparing diuretics** — S21
- **directly reflecting the percentage normally reabsorbed at each nephron segment in Figure 2.** This is why loops are reached for first when potent natriuresis is the goal. S21; S21 (notes)

- **TB** Potassium-sparing diuretics fit mineralocorticoid excess: primary (Conn syndrome) or secondary hyperaldosteronism (heart failure, cirrhosis, nephrotic syndrome). p. 274
- **TB** Hepatic cirrhosis: resistant to loops but unusually responsive to spironolactone; aggressive diuresis risks hepatorenal syndrome and encephalopathy. p. 279
- **TB** Renal & cardiac protection: aldosterone antagonists are cardioprotective and lower albuminuria; finerenone slowed CKD in type 2 diabetes. p. 281
- **TB** Idiopathic edema (women 20–30): rule out surreptitious diuretic use; prefer moderate salt restriction. p. 279

## ADVERSE EFFECTS (BOXED WARNINGS FIRST)

- **Primary adverse effect across the class: hyperkalemia — the risk rises sharply when combined with ACE inhibitors, ARBs, or in renal impairment (a direct preview of Session 2's cardiorenal drug-interaction content).** S17; SG U5-R5
- Spironolactone also has antiandrogen effects (gynecomastia, menstrual irregularities) that eplerenone largely avoids. S17
- Acetazolamide causes a mild metabolic acidosis as a predictable adverse effect, plus potassium loss; watch for kidney stones. S19; S20

## CONTRAINDICATIONS

- Mannitol requires intact renal perfusion to be effective. S19; S20 (notes)
- Eplerenone: avoid with potassium above 5.5 mEq/L, CrCl below 30 mL/min, or strong CYP3A4 inhibitors (such as ketoconazole). S33

- Mannitol expands extracellular volume — can worsen volume overload if used inappropriately; avoid it in heart failure and volume depletion. S19; S20
- **TB** $K^+$-sparing agents can cause hyperchloremic metabolic acidosis (like type IV RTA); triamterene can cause kidney stones and, with indomethacin, acute renal failure. p. 275
- **TB** Acetazolamide: drowsiness, paresthesias, calcium phosphate stones, hyperammonemia in cirrhosis; monitor potassium and phosphorus. p. 268
- **TB** Mannitol: early ECF expansion and hyponatremia, later dehydration, hypernatremia and hyperkalemia; AKI in 6–7% of patients. p. 276

## DRUG INTERACTIONS

- Hyperkalemia risk rises with ACE inhibitors, ARBs, or CKD — both block aldosterone's effect, so the classes are combined carefully. S18; S18 (notes)
- Flag the hyperkalemia-with-ACEi/ARB interaction as a preview: this exact combination becomes a major theme in Session 2 when RAAS blockers are introduced. S17 (notes)
- Potassium supplements and trimethoprim: hyperkalemia with potassium-sparing diuretics. S38; S38 (notes)
- Sulfonamide allergy: caution with loops, thiazides, acetazolamide. Metformin, vancomycin, DOACs: renally cleared; need dose adjustment. S38; S38 (notes)
- **TB** Strong CYP3A4 inhibitors (erythromycin, fluconazole, diltiazem, grapefruit juice) raise eplerenone but not spironolactone levels. p. 275
- **TB** RALES showed spironolactone helps severe heart failure on an ACE inhibitor — afterward hyperkalemia admissions and deaths rose. p. 275

## PHARMACOKINETICS

- **TB** Acetazolamide diuresis fades within 2–3 days (reduce dose in renal insufficiency); mannitol must be given IV — orally it causes osmotic diarrhea. p. 268; p. 275

### MONITORING

- Spironolactone: monitor potassium and creatinine within 1 week; life-threatening hyperkalemia, especially in CKD or with ACE inhibitor/ARB. S32
- Mannitol: monitor osmolality and electrolytes. S37

### CLINICAL PEARL / WHY

- Case: a patient with stage 3 chronic kidney disease and heart failure is taking an ACE inhibitor and is started on spironolactone for additional heart-failure benefit; two weeks later a routine basic metabolic panel shows potassium 6.1 mEq/L. S23
- **What is the mechanistic explanation for this finding, and what does it suggest about future drug combinations in this unit?** S23; S23 (notes)
- **Both the ACE inhibitor (by reducing aldosterone) and spironolactone (by directly blocking the aldosterone receptor) independently reduce potassium excretion — combined, and in the setting of CKD (which already impairs potassium excretion), the hyperkalemia risk compounds.** S23
- **This is the same additive-risk logic that will reappear as the "triple whammy" in Session 2.** The ACEi/aldosterone-antagonist pair is one of three “whammies” formalized next session. S23; S23 (notes)
- Recap: diuretic classes are organized by nephron site (Figure 2): loop diuretics (thick ascending limb) are most potent, followed by thiazides (distal tubule), then potassium-sparing and carbonic anhydrase inhibitors. S25
- Recap: loop and thiazide diuretics have opposite effects on calcium excretion (Figure 3); potassium-sparing diuretics carry a hyperkalemia risk that compounds with ACE inhibitors or ARBs. S25
- Session 2 (Thursday, November 19) continues with antihypertensive therapy in chronic kidney disease and the cardiovascular-renal overlap, including vasodilators and angina pharmacology — RAAS blockers and the triple whammy first. S25; S25 (notes)
- **TB** SGLT2 inhibitors block proximal glucose and $Na^+$ reabsorption; main indication now cardiorenal protection. Adverse reactions: sixfold genital fungal infections in women, initial GFR dip. p. 268; p. 269
- **TB** Antidiuretic hormone antagonists (vaptans): oral tolvaptan (V2), IV conivaptan (V1a and V2) for SIADH when water restriction fails; watch sodium and liver tests. p. 276; p. 277

**Step by step**

### The hyperkalemia setup S23; S32

CKD already limits $K^+$ excretion → ACE inhibitor lowers aldosterone →

Spironolactone blocks its receptor → $K^+$ rises (6.1 mEq/L) →

Check $K^+$ and creatinine within 1 week

**Figure 5 · Collecting duct: aldosterone blockers vs. ENaC blockers.** Schematic (illustrative) — Figure 5. Collecting duct: aldosterone-receptor blockers and ENaC blockers act at different steps (illustrative schematic). One blocks the hormone's receptor, the other blocks the channel the hormone turns on; both raise potassium. S18; S18 (notes)

**Figure 7 · Diuretic class comparison: natriuretic potency.** Original illustrative comparison (illustrative values, not from Katzung): loops block about 25% of filtered sodium reabsorption, thiazides about 6%, carbonic anhydrase inhibitors about 4%, potassium-sparing diuretics about 2% — the same nephron-segment logic as Figure 2, now quantified. S21; S21 (notes)

## Diuretic classes at a glance (Table 1)

| Class | Example agent(s) | Key teaching point |
|---|---|---|
| **Loop** | Furosemide, bumetanide, torsemide | Most potent; blocks NKCC2; ↑ $Ca^{2+}$ excretion; ototoxicity risk |
| **Thiazide** | Hydrochlorothiazide, chlorthalidone | First-line for HTN; ↓ $Ca^{2+}$ excretion; loses efficacy at low GFR |
| **$K^+$-sparing** | Spironolactone, eplerenone, amiloride, triamterene | Hyperkalemia risk, especially with ACEi/ARB |
| **CAI / Osmotic** | Acetazolamide / mannitol | Niche uses: glaucoma, altitude sickness / ↑ ICP, ↑ IOP |

How to tell them apart. Sources: S22; S22 (notes)

## Figure 6 · Proximal tubule: carbonic anhydrase inhibition and osmotic holding of water (Schematic (illustrative))

| | Acetazolamide | Mannitol |
|---|---|---|
| **How** | Blocks carbonic anhydrase → less $HCO_3^-$ reabsorbed | Filtered, not reabsorbed → water held in lumen |

| | Acetazolamide | Mannitol |
|---|---|---|
| **Result** | Bicarbonate diuresis, metabolic acidosis | Osmotic diuresis |
| **Uses** | Glaucoma, altitude sickness, raised intracranial pressure | Raised ICP or IOP |
| **Watch for** | Metabolic acidosis, potassium loss, kidney stones | Expands extracellular volume; avoid in heart failure and volume depletion |

How to tell them apart. Sources: S20; S20 (notes)

## Drug names reveal the class (S24)

| Stem | Class | Examples | Memory aid |
|---|---|---|---|
| **-semide** | Loop | Furosemide, torsemide | Works at the thick ascending limb; strongest class |
| **-tanide** | Loop | Bumetanide | About 40× furosemide potency, mg for mg |
| **-thiazide** | Thiazide | Hydrochlorothiazide | Acts at the distal convoluted tubule |
| **-thalidone** | Thiazide-like | Chlorthalidone | Same site, longer action |
| **-onolactone / -erenone** | Aldosterone antagonist | Spironolactone, eplerenone | Blocks the mineralocorticoid receptor |
| **-ride** | ENaC blocker | Amiloride | Blocks the sodium channel directly |
| **-zolamide** | Carbonic anhydrase inhibitor | Acetazolamide | Proximal tubule; bicarbonate diuresis |
| **mannitol / -itol** | Osmotic diuretic | Mannitol | Filtered, holds water in the lumen |

How to tell them apart. Sources: S24; S24 (notes)

## Supporting drugs at a glance (S38)

| Drug(s) | Why it matters here |
|---|---|
| **ACE inhibitors and ARBs** | Hyperkalemia with potassium-sparing diuretics |
| **NSAIDs** | Blunt diuretic effect; raise AKI risk |
| **Lithium** | Levels rise with thiazides |
| **Digoxin** | Hypokalemia from loops increases toxicity |
| **Aminoglycosides** | Additive ototoxicity with loops |
| **Potassium supplements and trimethoprim** | Hyperkalemia with potassium-sparing diuretics |
| **Sulfonamide allergy** | Caution with loops, thiazides, acetazolamide |

| Drug(s) | Why it matters here |
|---|---|
| **Metformin, vancomycin, DOACs** | Renally cleared; need dose adjustment |

How to tell them apart. Sources: S38; S38 (notes)

## Potassium-sparing profiles: formulation, dose, indications

| Drug | Class | Formulation | Adult dose | Indications |
|---|---|---|---|---|
| **Spironolactone (Aldactone)** | **Aldosterone receptor antagonist (potassium-sparing)** | Tablets 25/50/100 mg; oral suspension | Hypertension: 25-100 mg daily; heart failure: 12.5-25 mg daily; primary hyperaldosteronism: 100-400 mg daily | Heart failure, resistant hypertension, primary hyperaldosteronism, edema, ascites |
| **Eplerenone (Inspra)** | **Selective aldosterone receptor antagonist** | Tablets 25/50 mg | Heart failure post-MI: 25 mg daily, increase to 50 mg; hypertension: 50 mg daily, may increase to BID | Heart failure after myocardial infarction, hypertension |
| **Amiloride (Midamor)** | **ENaC blocker (potassium-sparing)** | Tablets 5 mg | 5 mg PO daily, may increase to 10 mg (maximum 20 mg/day) | Adjunct to loop or thiazide to preserve potassium, lithium-induced nephrogenic diabetes insipidus |
| **Triamterene (Dyrenium; Maxzide and Dyazide combinations)** | **ENaC blocker (potassium-sparing)** | Capsules 50/100 mg; combination tablets with hydrochlorothiazide (37.5/25 and 75/50 mg) | Alone: 100 mg PO BID (maximum 300 mg/day); combination: 37.5/25 to 75/50 mg daily | Edema, hypertension, hypokalemia prevention |

How to tell them apart. Sources: S32; S33; S34; S35

## Potassium-sparing profiles: adverse effects, contraindications, toxicity

| Drug | Adverse effects | Contraindications | Toxicity / overdose |
|---|---|---|---|
| **Spironolactone (Aldactone)** | Hyperkalemia, gynecomastia, menstrual irregularity, breast tenderness, GI upset | Hyperkalemia, Addison disease, anuria; avoid with potassium supplements and caution with ACE inhibitors or ARBs | Life-threatening hyperkalemia, especially in CKD or with ACE inhibitor/ARB; monitor potassium and creatinine within 1 week |

| Drug | Adverse effects | Contraindications | Toxicity / overdose |
|---|---|---|---|
| **Eplerenone (Inspra)** | Hyperkalemia, dizziness, hypertriglyceridemia, fewer antiandrogen effects | Potassium above 5.5 mEq/L, CrCl below 30 mL/min, strong CYP3A4 inhibitors (such as ketoconazole) | Hyperkalemia; levels rise sharply with CYP3A4 inhibitors; monitor potassium and creatinine |
| **Amiloride (Midamor)** | Hyperkalemia, nausea, headache, weakness | Potassium above 5.5 mEq/L, anuria, significant renal impairment, concomitant potassium supplements | Hyperkalemia with arrhythmias, especially in diabetes and CKD; monitor potassium |
| **Triamterene (Dyrenium; Maxzide and Dyazide combinations)** | Hyperkalemia, nausea, kidney stones, blue urine, folate antagonism | Hyperkalemia, severe renal impairment, anuria, concomitant potassium supplements | Hyperkalemia, nephrolithiasis, AKI with NSAIDs; monitor potassium and creatinine |

How to tell them apart. Sources: S32; S33; S34; S35

## Acetazolamide and mannitol profiles: formulation, dose, indications

| Drug | Class | Formulation | Adult dose | Indications |
|---|---|---|---|---|
| **Acetazolamide (Diamox)** | **Carbonic anhydrase inhibitor** | Tablets 125/250 mg; extended-release capsules 500 mg; IV 500 mg | Glaucoma: 250 mg PO 1-4 times daily; altitude sickness: 125-250 mg BID starting a day before ascent; edema: 250-375 mg daily | Glaucoma, altitude sickness prophylaxis, metabolic alkalosis, idiopathic intracranial hypertension |
| **Mannitol (Osmitrol)** | **Osmotic diuretic** | IV solutions 10%, 15%, 20% and 25% | Raised ICP or IOP: 0.25-1 g/kg IV over 30-60 minutes; repeat 0.25-0.5 g/kg every 4-8 hours as needed | Raised intracranial pressure, raised intraocular pressure, oliguria prevention in selected cases |

How to tell them apart. Sources: S36; S37

## Acetazolamide and mannitol profiles: adverse effects, contraindications, toxicity

| Drug | Adverse effects | Contraindications | Toxicity / overdose |
|---|---|---|---|
| **Acetazolamide (Diamox)** | Metabolic acidosis, paresthesias, hypokalemia, kidney stones, taste changes | Sulfonamide allergy, severe renal or hepatic disease, hyponatremia, hypokalemia, adrenal failure | Severe metabolic acidosis, hypokalemia, confusion, bone marrow suppression (rare) |
| **Mannitol (Osmitrol)** | Fluid and electrolyte shifts, headache, nausea, thirst, hyponatremia | Anuria, severe dehydration, active intracranial bleeding, severe pulmonary edema, heart failure | Volume overload with pulmonary edema, hyperosmolality, AKI, hypo- or hypernatremia; monitor osmolality and electrolytes |

How to tell them apart. Sources: S36; S37

## Other renal agents in the reading (textbook)

| Drug / group | Mechanism | Point |
|---|---|---|
| **SGLT2 inhibitors (dapagliflozin, empagliflozin, canagliflozin)** | Block proximal $Na^+$/glucose cotransport | Cardio- and renoprotective; genital fungal infections |
| **Finerenone** | Nonsteroidal mineralocorticoid antagonist | Less hyperkalemia (2.8%); slows diabetic CKD |
| **Metolazone** | Thiazide-like | Added to loops for synergy; start 2.5 mg weekly |
| **Ethacrynic acid** | Loop, not a sulfonamide | Option for true sulfa allergy |
| **Tolvaptan / conivaptan** | V2 / V1a+V2 ADH antagonists | SIADH, heart-failure hyponatremia; liver tests |
| **Dorzolamide, brinzolamide** | Topical carbonic anhydrase inhibitors | Glaucoma without systemic effects |

How to tell them apart. Sources: p. 267; p. 268; p. 273; p. 276; p. 277; p. 281

**SLIDE DISCREPANCY**

- **Mannitol dose for raised ICP/IOP**
- Slides: 0.25–1 g/kg IV over 30–60 minutes. S37
- Textbook: 1–2 g/kg IV. p. 275

**WHY IT HAPPENS**

- Why does ACE inhibitor + spironolactone raise potassium so much? Both cut aldosterone's effect — one lowers the hormone, the other blocks its receptor — and CKD already limits $K^+$ excretion. S23

**MNEMONIC (MEMORY AID)**

- Stems: **-semide**, **-tanide** loop · **-thiazide**, **-thalidone** distal · **-onolactone / -erenone**, **-ride** collecting duct · **-zolamide** proximal · **mannitol / -itol** osmotic. Read the **Stem**, name the **Class**, recall the **Examples** and the **Memory aid**. S24; S24 (notes)

# Cram sheet

Every term, mechanism, drug, and number. Hide the answers and quiz yourself; tap a blurred answer to peek.

| Prompt | Answer |
|---|---|
| **Falling GFR** | Accumulation → reduce dose, extend interval, or both S5; S6; SG U5-R1 |
| **Loading vs. maintenance** | Loading often unchanged; maintenance adjusted S8 |
| **$Na^+$ reabsorbed by segment** | PT ~65% · TAL ~25% · DCT 5–8% · CD 2–3% S7 (notes) |
| **Loop target** | NKCC2, thick ascending limb S10 |
| **Loop AEs** | ↓K, ↓Mg, alkalosis, ototoxicity, hyperuricemia S10 |
| **Thiazide target** | NCC, distal convoluted tubule S12 |
| **Thiazide AEs** | ↓Na (older adults), ↓K, ↑glucose, ↑uric acid, ↑Ca S12 |
| **Calcium** | Loops ↑ excretion · thiazides ↓ excretion S14; SG U5-R3 |
| **Thiazide cutoff** | GFR < ~30 → switch to a loop S15; SG U5-R4 |
| **Chlorthalidone** | Longer half-life (40–60 h), 24-hour BP control S12; S31 |
| **Bumetanide** | About 40× furosemide mg for mg S28 |
| **$K^+$-sparing subgroups** | Spironolactone, eplerenone (receptor) · amiloride, triamterene (ENaC) S17; SG U5-R5 |
| **Spironolactone vs. eplerenone** | Gynecomastia vs. fewer antiandrogen effects; eplerenone + CYP3A4 inhibitors S17; S33 |
| **Hyperkalemia setup** | $K^+$-sparing + ACEi/ARB + CKD S17; S23 |
| **Acetazolamide** | Glaucoma, altitude sickness, alkalosis; causes acidosis S19; SG U5-R6 |
| **Mannitol** | Raised ICP/IOP; avoid in heart failure S19; S20 |
| **Digoxin + loop** | Hypokalemia → digoxin toxicity S38 |
| **Lithium + thiazide** | Lithium levels rise S38 |
| **Maintenance vs. loading dose** | Maintenance = CL × target · loading = V × target p. 51; p. 52 |
| **Steady state** | About four half-lives p. 47 |
| **Hypercalcemia regimen** | Normal saline + IV furosemide 80–120 mg p. 280 |
| **Loop resistance** | Add metolazone (loop + thiazide synergy); watch $K^+$ p. 277 |

| Prompt | Answer |
| --- | --- |
| **Cirrhotic ascites** | Spironolactone (loops resisted) p. 279 |
| **Eplerenone + CYP3A4 inhibitors** | Levels rise; spironolactone unaffected p. 275 |
| **SIADH, water restriction failed** | Tolvaptan (oral V2) or conivaptan (IV) p. 276 |
| **SGLT2 inhibitors** | Cardiorenal protection; genital fungal infections p. 268; p. 269 |

# Self-test

First, the lecture's own 0 knowledge checks — exactly as written. Then original PANCE-style questions. Tap an answer to reveal the explanation.

## From the lecture

## Original questions

Q1 · Mechanism · Renal Dosing & Diuretic Sites

**Why does a falling GFR change how a renally cleared drug must be dosed?**

- **A** Absorption from the gut falls with GFR
- **B** Only the parent drug matters, not metabolites
- **C** The loading dose must always be doubled
- **D** The drug accumulates unless dose or interval is adjusted
- **E** The drug is cleared faster, so doses rise

**ANSWER: D**

Reduced GFR causes a renally cleared drug to accumulate if unadjusted; reduce the dose, extend the interval, or both (S5; SG U5-R1). S5; S8; SG U5-R1

- **A.** GFR affects elimination, not absorption (S5).
- **B.** Active metabolites matter too (S8).
- **C.** Loading doses are often unchanged (S8).
- **E.** Clearance falls, not rises (S5).

Q2 · Clinical vignette · Renal Dosing & Diuretic Sites

**An 80-year-old has pneumonia and dehydration. She takes metformin and a DOAC. What is the best dosing habit?**

- **A** Stop checking renal function in older adults
- **B** Keep last year's doses since CrCl is stable
- **C** Re-estimate CrCl now and adjust renally cleared drugs
- **D** Double the loading dose of both drugs
- **E** Adjust only the drugs with toxic metabolites

**ANSWER: C**

Acute illness and dehydration can drop GFR quickly; estimate CrCl or eGFR and reassess renally cleared drugs (S8). S5; S8

- **A.** Older adults are the priority (S8).
- **B.** GFR can fall quickly (S8).
- **D.** Loading doses are often unchanged (S8).
- **E.** Parent drugs count too (S8).

Q3 · Mechanism · Renal Dosing & Diuretic Sites

**Roughly what share of filtered sodium does each segment reabsorb, from proximal tubule to collecting duct?**

- **A** About 5–8%, 25%, 65%, and 2–3%
- **B** About equal at 25% in every segment
- **C** About 25%, 65%, 2–3%, and 5–8%
- **D** About 65%, 25%, 5–8%, and 2–3%
- **E** About 2–3%, 5–8%, 25%, and 65%

**ANSWER: D**

Proximal tubule ~65%, thick ascending limb ~25%, distal tubule 5–8%, collecting duct 2–3% (S7 notes). S7; S7 (notes)

- **A.** The proximal tubule reabsorbs most (S7).
- **B.** Shares differ by segment (S7).
- **C.** Proximal and loop are swapped (S7).
- **E.** Reversed (S7).

Q4 · Mechanism · Loop & Thiazide Diuretics

**Why do loop diuretics increase calcium excretion while thiazides decrease it?**

- **A** Loops collapse lumen potential; thiazides aid NCX
- **B** Thiazides act on NKCC2; loops act on NCC
- **C** Loops block calcium channels; thiazides open them
- **D** Both raise calcium excretion equally
- **E** Loops raise PTH; thiazides suppress PTH

**ANSWER: A**

Loops collapse the lumen-positive potential driving paracellular $Ca^{2+}$/$Mg^{2+}$ reabsorption; thiazides lower cell $Na^{+}$ so NCX reabsorbs more $Ca^{2+}$ (S11; S13; SG U5-R3). S11; S13; SG U5-R3

- **B.** Sites are swapped (S10; S12).
- **C.** Not the mechanism (S11; S13).
- **D.** Effects are opposite (S14).
- **E.** Not the taught mechanism (S11).

Q5 · Clinical vignette · Loop & Thiazide Diuretics

**A patient has recurrent calcium kidney stones and hypertension with normal renal function. Which diuretic fits best?**

- **A** Mannitol (osmotic diuretic)
- **B** Furosemide (loop diuretic)
- **C** Hydrochlorothiazide (thiazide)
- **D** Acetazolamide (CA inhibitor)
- **E** Bumetanide (loop diuretic)

**ANSWER: C**

Thiazides decrease calcium excretion, useful in recurrent calcium stones; first-line HTN with preserved renal function (S12; S30). S12; S30

- **A.** IV osmotic agent for ICP/IOP (S37).
- **B.** Increases calcium excretion (S10).
- **D.** Kidney stones are an adverse effect (S36).
- **E.** Loop; raises urinary calcium (S10).

Q6 · Clinical vignette · Loop & Thiazide Diuretics

**A patient with heart failure on digoxin starts high-dose furosemide. Which lab change most raises digoxin toxicity risk?**

- **A** Hypokalemia
- **B** Metabolic acidosis
- **C** Hypercalcemia
- **D** Hyperkalemia
- **E** Hyponatremia

**ANSWER: A**

Hypokalemia from loops increases digoxin toxicity (S38). S10; S38

- **B.** Loops cause alkalosis (S10).
- **C.** Loops increase calcium excretion (S10).
- **D.** Loops lower potassium (S10).
- **E.** Not the digoxin link (S38).

Q7 · Mechanism · Loop & Thiazide Diuretics

**Which adverse effect is most characteristic of loop diuretics given rapidly IV at high doses?**

- **A** Metabolic acidosis
- **B** Ototoxicity
- **C** Hypercalcemia
- **D** Gynecomastia
- **E** Hyperkalemia

**ANSWER: B**

Ototoxicity, especially with rapid IV administration or high doses; give IV furosemide no faster than 4 mg/min (S10; S27). S10; S27

- **A.** Acetazolamide (S19).
- **C.** Thiazides (S12).
- **D.** Spironolactone (S17).
- **E.** Potassium-sparing class (S17).

Q8 · Clinical vignette · Loop & Thiazide Diuretics

**An older woman on hydrochlorothiazide is confused, with sodium 122 mEq/L. Which statement fits?**

- **A** Thiazide hyponatremia, especially in older adults
- **B** Thiazides raise sodium by blocking ADH
- **C** This is mannitol-type osmotic diuresis
- **D** Loop diuretics cause more hyponatremia than thiazides
- **E** Hyperkalemia explains the confusion

**ANSWER: A**

Hyponatremia can be significant with thiazides, especially in older adults (S12). S12; S30

- **B.** They lower sodium (S12).
- **C.** Not an osmotic agent (S19).
- **D.** Thiazides do (S12; p. 271).
- **E.** Thiazides lower potassium (S12).

Q9 · Clinical vignette · Loop & Thiazide Diuretics

**A patient with CKD on furosemide 80 mg twice daily has hydrochlorothiazide 25 mg added. Two weeks later he has lost 15 kg, BP is 91/58 and creatinine is 10.8. What happened?**

- **A** Ototoxicity from the furosemide
- **B** Thiazide failed, so fluid built up
- **C** Thiazide-induced hypercalcemia
- **D** Hyperkalemia from the thiazide
- **E** Loop–thiazide synergy caused hypovolemic AKI

**ANSWER: E**

Adding a thiazide to chronic loop therapy can cause dramatic diuresis, hypovolemia and AKI; monitor very closely, ideally inpatient (p. 277; p. 285). p. 261; p. 277; p. 285

- **A.** Not the picture (p. 285).
- **B.** He lost 15 kg (p. 285).
- **C.** Not the cause of AKI here (p. 285).
- **D.** Thiazides waste $K^+$ (p. 272).

Q10 · Mechanism · Loop & Thiazide Diuretics

**Why must saline be given with furosemide to treat hypercalcemia?**

- **A** Saline blocks NKCC2 with the loop
- **B** Saline prevents loop ototoxicity in the ear
- **C** Volume loss boosts proximal $Ca^{2+}$ reabsorption
- **D** Furosemide raises calcium without saline
- **E** Saline lowers PTH secretion directly

**ANSWER: C**

A loop alone causes volume contraction, which enhances proximal $Ca^{2+}$ reabsorption and stops the calcium diuresis; give saline with it (p. 280). p. 280

- **A.** Saline replaces volume (p. 280).
- **B.** Not the reason (p. 280).
- **D.** It lowers calcium (p. 280).
- **E.** Not the reason (p. 280).

Q11 · Mechanism · Potassium-Sparing, CAI & Osmotic Diuretics

**Which pairing correctly matches a potassium-sparing diuretic to its mechanism?**

- **A** Amiloride — inhibits carbonic anhydrase
- **B** Spironolactone — blocks ENaC directly
- **C** Triamterene — blocks aldosterone receptor
- **D** Amiloride — blocks ENaC directly
- **E** Eplerenone — blocks NKCC2

**ANSWER: D**

Amiloride and triamterene block ENaC; spironolactone and eplerenone block the mineralocorticoid receptor (S17–S18). S17; S18; SG U5-R5

- **A.** That is acetazolamide (S19).
- **B.** Receptor antagonist (S18).
- **C.** ENaC blocker (S17).
- **E.** Receptor antagonist (S17).

Q12 · Clinical vignette · Potassium-Sparing, CAI & Osmotic Diuretics

**A man needs an aldosterone antagonist for heart failure but developed painful gynecomastia on one before. Which fits?**

- **A** Acetazolamide (CA inhibitor)
- **B** Eplerenone (selective blocker)
- **C** Amiloride (ENaC blocker)
- **D** Hydrochlorothiazide (thiazide)
- **E** Spironolactone (higher dose)

**ANSWER: B**

Spironolactone has antiandrogen effects (gynecomastia) that eplerenone largely avoids (S17; S33). S17; S33

- **A.** Carbonic anhydrase inhibitor (S19).
- **C.** Not an aldosterone receptor blocker (S17).
- **D.** Not an aldosterone antagonist (S12).
- **E.** Causes the gynecomastia (S17).

Q13 · Clinical vignette · Potassium-Sparing, CAI & Osmotic Diuretics

**A hiker plans a rapid ascent to high altitude next week. Which diuretic is used for prophylaxis?**

- **A** Spironolactone, daily for a week
- **B** Hydrochlorothiazide, once daily
- **C** Acetazolamide, from a day before
- **D** Furosemide, daily during ascent
- **E** Mannitol, IV at the summit

**ANSWER: C**

Acetazolamide is used for altitude sickness prophylaxis, 125–250 mg BID starting a day before ascent (S19; S36). S19; S36; SG U5-R6

- **A.** Heart failure, hyperaldosteronism (S32).
- **B.** Hypertension (S12).
- **D.** Edema, not altitude (S10).
- **E.** IV for raised ICP/IOP (S19).

Q14 · Mechanism · Potassium-Sparing, CAI & Osmotic Diuretics

**How does mannitol produce diuresis?**

- **A** It is filtered but not reabsorbed, holding water in the lumen
- **B** It antagonizes aldosterone at its receptor
- **C** It inhibits carbonic anhydrase in the proximal tubule
- **D** It blocks NKCC2 in the thick ascending limb
- **E** It blocks ENaC in the collecting duct

**ANSWER: A**

Mannitol is an inert, filtered but poorly reabsorbed solute that holds water in the tubule lumen osmotically (S19–S20). S19; S20

- **B.** That is spironolactone (S18).
- **C.** That is acetazolamide (S19).
- **D.** That is a loop (S10).
- **E.** That is amiloride (S17).

Q15 · Clinical vignette · Potassium-Sparing, CAI & Osmotic Diuretics

**A patient with decompensated heart failure and raised intracranial pressure is being considered for mannitol. What is the concern?**

- **A** Mannitol is safe because it is inert
- **B** Mannitol expands ECF volume and can worsen heart failure
- **C** Mannitol has no effect on fluid status
- **D** Mannitol causes severe hyperkalemia via ENaC
- **E** Mannitol causes ototoxicity like loops

**ANSWER: B**

Mannitol expands extracellular volume; avoid it in heart failure and volume depletion; needs intact renal perfusion (S19–S20; S37). S19; S20; S37

- **A.** Inert, but volume overload (S19).
- **C.** It shifts fluid (S20).
- **D.** Not its mechanism (S20).
- **E.** That is loops (S10).

Q16 · Mechanism · Potassium-Sparing, CAI & Osmotic Diuretics

**Which acid–base change is a predictable adverse effect of acetazolamide?**

- **A** Respiratory acidosis
- **B** Respiratory alkalosis
- **C** Metabolic alkalosis
- **D** No acid–base change
- **E** Metabolic acidosis

**ANSWER: E**

Blocking carbonic anhydrase causes bicarbonate diuresis and a mild metabolic acidosis (S19–S20). S19; S20

- **A.** Not the drug effect (S20).
- **B.** Acetazolamide counters it at altitude (p. 267).
- **C.** Loops and thiazides (S10).
- **D.** Acidosis is predictable (S19).

Q17 · Clinical vignette · Potassium-Sparing, CAI & Osmotic Diuretics

**A patient with SIADH stays hyponatremic despite fluid restriction at home. Which drug does the textbook use next?**

- **A** Desmopressin (ADH agonist)
- **B** Mannitol (given orally)
- **C** Hydrochlorothiazide (thiazide)
- **D** Tolvaptan (oral V2 blocker)
- **E** Spironolactone (MR blocker)

**ANSWER: D**

ADH antagonists are used for SIADH when water restriction fails; tolvaptan is oral, conivaptan IV (p. 276). p. 276

- **A.** An ADH agonist (p. 280).
- **B.** Oral mannitol causes diarrhea (p. 275).
- **C.** Thiazides cause hyponatremia (p. 272).
- **E.** Not an ADH antagonist (p. 273).

# Objectives checklist

Tick only what you could teach out loud. Saved on this device.

- [ ] **IO5.1** Identify major pharmacologic classes used in the management of renal conditions and describe their mechanisms of action.
- [ ] **IO5.2** Describe the indications, contraindications, adverse effects, and clinically significant interactions associated with medications used to treat renal conditions.
- [ ] **IO5.3** Compare pharmacologic agents within and across therapeutic classes to distinguish clinically relevant differences in efficacy, safety, and use.
- [ ] **IO5.4** Select appropriate pharmacologic therapy for common renal conditions based on clinical presentation and patient-specific factors.
- [ ] **IO5.5** Identify appropriate dosing, administration, and monitoring considerations for medications used in renal conditions.
- [ ] **IO5.6** Explain important medication-use and safety information to patients receiving pharmacotherapy for renal.
- [ ] **SO1** Identify the major pharmacologic classes used in the management of renal conditions — diuretics — and describe their mechanisms of action at specific nephron sites.
- [ ] **SO2** Apply basic pharmacokinetic principles to drug dosing in renal impairment, including creatinine clearance/GFR-based dose and interval adjustment.
- [ ] **SO3** Describe the indications, contraindications, adverse effects, and clinically significant interactions of loop, thiazide, potassium-sparing, carbonic anhydrase inhibitor, and osmotic diuretics.
- [ ] **SO4** Compare diuretic classes to distinguish clinically relevant differences in efficacy, electrolyte effects, and clinical use.
- [ ] **SO5** Identify dosing, administration, and monitoring considerations, and explain key medication-safety counseling points for patients on diuretic therapy.